NAME: ____ CLASS: __________

DIOCESE OF MZUZU EXAMINATIONS

2019 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

AGRICULTURE

Friday, 29 March

Subject Number: M012/1
Time Allowed: 2 hours
(7.30-09.30am)

PAPER 1
(100 Marks)
(Theory)

Instructions:

1. This paper contains **15 pages**. Please check.
2. Answer all **ten** questions in **section A** and all **three** questions in **section B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Answer all questions in the spaces provided.
5. Write your **name** and **class** on top of every page.
6. In the table provided on this page, tick against the question number you have answered.
7. Hand in your paper to the invigilator at the end.

Question Number	Tick if Answered	Do not write in these margins	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			

Section A (70 marks)

1. a. Mention any **two** ways how soil fertility is lost in farms.

____

____

(2 marks)

b. Explain any **two** ways in which soil fertility can be maintained.

____

____

____

____

(4 marks)

c. State any **two** effects of soil degradation on crop production.

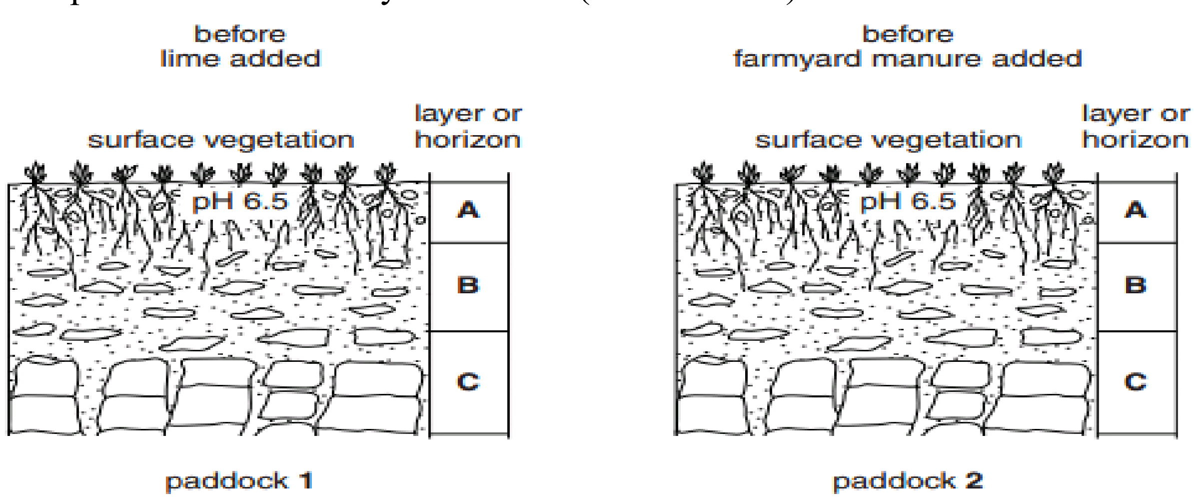
____

____

(2 marks)

d. **Figure 1** shows the soil profiles in two paddocks. The surface of each paddock was treated as follows:

- paddock 1 with lime,
- paddock 2 with farmyard manure (kraal manure)



Suggest a pH value for **layer A** in both paddocks **three** weeks after treatment. Give a reason for each of your answers.

i. Paddock 1 pH value

____

(1 mark)

Reason

____

____

(1 mark)

ii. Paddock 2 pH value

____

(1 mark)

Reason

____

____

(1 mark)

2. **Table 1** shows an experiment to find the effect of adding different fertilizers to the yield of potato and wheat crops. Use it to answer the questions that follow.

plot	fertiliser added /kg per hectare			crop yield /tonnes per hectare	
	N	P	K	wheat	potatoes
A	0	0	0	1.7	9.5
B	96	0	0	3.7	8.3
C	0	77	107	2.0	16.7
D	96	77	107	6.6	38.6

a. (i) Which plot was the control?

____

(1 mark)

(ii) Give a reason for your answer to **a (i)**.

____

____

(1 mark)

b. What is the effect of adding only N fertilizer on the yield of potatoes?

____

____

(1 mark)

c. Use the data in the table to describe the effect on the yield of wheat of adding P and K

i. without N

____

____

(1 mark)

ii. with N

____

____

(1 mark)

d. Explain any **one** way in which this experiment is important in crop production.

____

____

(2 marks)

3. **Table 2** shows information on green maize and groundnuts. Use it to answer the questions that follow.

Item	**Groundnuts**	**Green maize**
Groundnuts: 100 bags at K3,000 per bag Green Maize: 10,000 cobs at K50 per cob	300, 000	500, 000
Total Revenue	300, 000	500, 000
Total Variable costs	25, 000	35, 000
Total fixed costs	15, 000	20, 000

a. Calculate each of the following:

i. Break-even quantity for the Green Maize

(3 marks)

ii. Break-even price for the Groundnuts.

(3 marks)

b. State any **two** ways in which calculation of break-even budget is important in farm business management.

(2 marks)

4. **Figure 2** is a diagram of a boy and a girl irrigating vegetables. Use it to answer the questions that follow.

Figure 2

a. Identify the type of irrigation system shown in **Figure 2.**

(1 mark)

b. State any **one** disadvantage of this irrigation system.

(1 mark)

c. Explain any **one** way in which this agricultural technology can help in mitigation of climate change in agricultural production.

____

____

____

(2 marks)

5. **Table 3** shows price of soya beans and the quantity demanded and supplied at Jenda Market. Use it to answer the questions that follow.

Price (MK)	**Quantity Demanded (kg)**	**Quantity Supplied (kg)**
50	500	150
100	400	200
150	300	250
200	200	300
250	100	350

a. Taking a scale of 2cm to represent 100 units in the *x*-axis and 2cm to represent 50 units in the *y*-axis, draw the graph of price of soya beans against the quantity demanded and quantity supplied on the same pairs of axes.

(5 marks)

b. Identify the equilibrium price from the graph.

____

(1 mark)

c. Explain any **one** effect of charging the price of Soya beans above the price mentioned in **(b)** above.

____

____

(2 marks)

6. a. What do you understand by each of the following terms as used in mushroom production?

   i. Spawning

   ____

   (1 mark)

   ii. Substrate

   ____

   (1 mark)

   b. Explain any **one** way in which casing soil is useful in mushroom production.

   ____

   ____

   (2 marks)

   c. State any **two** activities involved in mushroom fruiting management.

   ____

   ____

   (2 marks)

7. A farmer harvested his Groundnut from a field, before storing them, the groundnuts were dried on the sunlight.

   a. Give any **two** benefits of this activity.

   ____

   ____

   (2 marks)

   b. Mention **two** other processes to be done to ground nuts before they are stored**.**

   ____

   ____

(2 marks)

8. A farmer applied farm yard manure to a Vegetable crop.

   i. State the type of fertilizer mentioned above.

   ____

   (1 mark)

   ii. Explain any **one** advantage and any **one** disadvantage of this type of fertilizer.

   Advantage

   ____

   ____

   (2 marks)

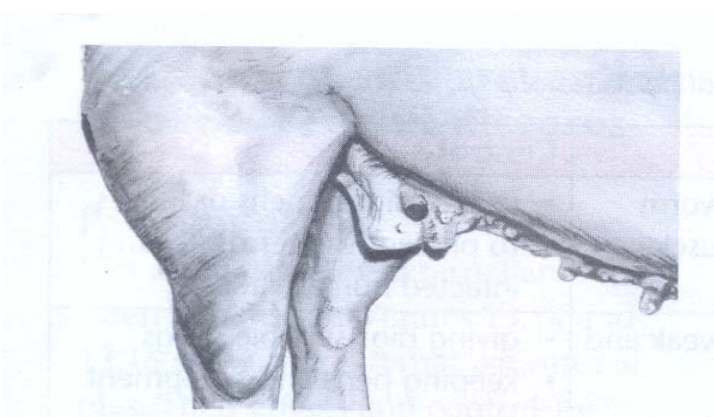
   Disadvantage

   ____

   ____

   (2 marks)

9. **Figure 3** shows one of the most common diseases of pigs. Use it to answer the questions that follow.



   a. Identify the disease.

   ____

   (1 mark)

b. State any **two** symptoms of the disease mentioned in **(a)** above

____

____

(2 marks)

c. Give any **two** ways of controlling the disease named in **(a)** above.

____

____

(2 marks)

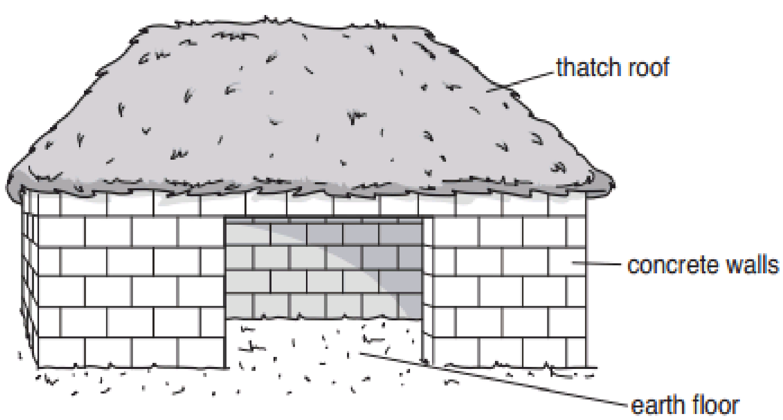
d. Explain any **one** importance of wallowing in pig production.

____

____

(2 marks)

10. a. **Figure 4** shows a house for a commercial farmer raising cattle. Use it to answer the questions that follow.



Give any **one** reason for each of the following construction choices

i. Concrete blocks are used for the walls;

____

____

(1 mark)

ii. Thatch is used for the roof;

____

____

(1 mark)

iii. Earth is used for the floor.

____

____

(1 mark)

b. (i) State any **three** characteristics of livestock to be selected for breeding.

____

____

____

____

(3 marks)

(ii) Give any **two** signs of heat of a nanny.

____

____

____

(2 marks)

(iii) Give any **one** class of livestock feed on a farm

____

____

(1 mark)

Section B (30 marks)

Answer all the three questions in this section

11. Explain any **five** ways in which agro-forestry reduces the effects of climate change.

(10 marks)

12. Explain any **five** factors that a farmer should consider when mechanizing a farm.

(10 marks)

13. Describe any **five** ways in which farmers in Malawi can adjust to risks and uncertainties.

(10 marks)

END OF THE QUESTION PAPER

NB: This paper contains 15 pages